PROJECT ON APPLICATION OF C

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CLASS: XII ROLL NO.: 10

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SUBJECT OF THE PROJECT: BISECTION METHOD.

BISECTION METHOD

The method is based on the theorem which states that "If a function $f(x)$ is continuous in the closed interval $[a,b]$ and $f(a)$ and $f(b)$ are of opposite signs[i.e. if $f(a)f(b)< 0$] then there exists at least one real root of $f(x)=0$, between a and b."

Let $f(a)$ and $f(b)$ be of opposite signs i.e. $f(a)$ $f(b)< 0$.

Let $\xi =(a+b)/ 2$, be the middle point of $[a,b]$.

Now, if $f(\xi)=0$, then ξ is a root of $f(x)=0$. If $f(\xi) \neq 0$, then either $f(a)\ f(\xi)< 0$ or $f(b)$ $f(\xi)< 0$

Application of C Programme

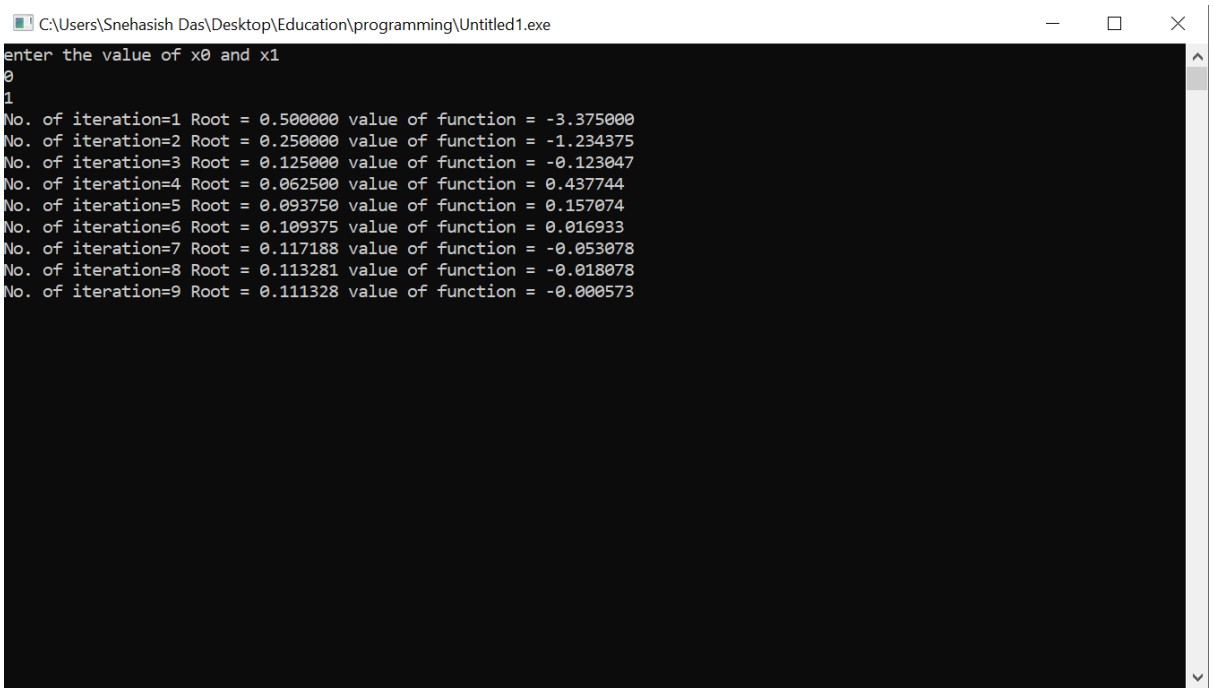
SOURCE CODE OF THE PROGRAM

```
//C PROJECT:

//BISECTION METHOD:

//FUNCTION F(X)= X^3-9X+1

#include<stdio.h>

#include<conio.h>

#include<math.h>

#define eps 0.001

#define f(x) x*x*x -9*x+1

void main()

{

float x0, x1,x2;

float f0,f1,f2;

int i=0;

clrscr();

printf("enter the value of x0 and x1\n");

scanf("%f%f",&x0,&x1);

do

{

f0= f(x0);

f1= f(x1);

x2=(x0+x1)/2;

f2=f(x2);

if(f0*f2 <0)

x1=x2;

else

x0=x2;
```

```
i++;

printf("No. of iteration=%d",i);

printf(" Root = %f",x2);

printf(" value of function = %f\n",f2);

}while(fabs(f2)>eps);

getch();

}
```

OUTPUT

BISECTION METHOD: F(X) =X^3-9X+1



ACKNOWLEDGEMENT

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I am indebted to Mr Chandan Kumar Das the Head Master of our school for his constant encouragement in my perseverance.

I would also like to convey my regards to my parents and my class mates for his day-to-day co-operation and guidance, which help me to complete the project in time.

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Signature of the Student *Signature of the Teacher*

CONCLUSION

I have tried my best to complete my Application C Programme project work successfully amidst amicable and ambience of constant co-operation &remarkable assistance. Varied tasks may be performed through various modules for easy accessibility. However, this recently developed system may not provide all sorts of facilities. Although there was no lack of endeavour on my part to develop this system but it is quite possible that there may be some potentials to develop it further which may due to lack of front-end application.

Inspite of all adversities I am elated to be able to complete these projects gathering huge experience in the process which will certainly help me in future.